

Nephrostomy

A nephrostomy lets urine (pee) drain from the kidney if there is a blockage in the urinary system.

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What is a nephrostomy?

A nephrostomy is a tube that lets urine drain from the kidney through an opening in the skin on the back.

A thin, flexible tube goes through the opening and into the kidney. This is called a nephrostomy tube or catheter. Urine drains through the tube into a bag outside your body. The bag has a tap so you can empty it. You may still pass some urine in a normal way even when you have a nephrostomy.

The urinary system

The urinary system is made up of the kidneys, the ureters, the bladder and the urethra.

The kidneys are at the back of the body, one on each side, just underneath the ribcage. The kidneys are bean-shaped. They are just above the level of the belly button. Kidneys filter your blood to get rid of waste products, and change them into urine (pee).

Urine travels from each kidney through a fine tube called a ureter to the bladder, where it is stored. The bladder is in the lower part of the tummy (abdomen). This area is called the pelvis.

When you are ready to pass urine, it leaves the bladder through a tube called the urethra. The urethra opens in front of the vagina, or at the tip of the penis.

Reasons for having a nephrostomy

You may need a nephrostomy if the cancer, or cancer treatment, affects one or both ureters. If a ureter becomes blocked, urine (pee) cannot flow from the kidney to the bladder. This causes urine to build up in the kidney. When this happens, the kidney may slowly stop working.

If this is not treated quickly it can make you feel very unwell. You may:

- have pain or discomfort in your back or side
- have a high temperature (fever)
- feel sick.

You may need 1 or 2 nephrostomy tubes, depending on whether one or both ureters are blocked. If you have 1 tube only, this is called a unilateral nephrostomy. If you have tubes in both ureters, this is called bilateral nephrostomies.

Cancers that start in the lower tummy (pelvis) are most likely to block the ureters. These include cancers of the:

- [bladder](#)
- [colon](#)
- [rectum](#)
- [cervix](#)
- [womb](#)
- [ovaries](#)
- [prostate](#).

Sometimes, a cancer that started in another part of the body spreads to the pelvis and forms a tumour. This can block 1 or both ureters. You usually have a scan to find out exactly where the blockage is.

How long do you need a nephrostomy tube for?

Your doctor will explain how long you are likely to have a nephrostomy tube. They may remove it if treatment for the cancer clears the blockage. Sometimes, the doctor puts a tube called a stent into the ureter. The stent holds the ureter open so it is no longer blocked. This means the nephrostomy tube can be removed.

If cancer treatment or a stent does not clear the blockage, the nephrostomy tube is left in. The doctor replaces it every 3 to 4 months, to stop it from blocking. Replacing a nephrostomy tube is a very quick procedure.

Having a nephrostomy placed

You usually have a local anaesthetic for a nephrostomy insertion. Most people only stay in hospital for a short time. A doctor usually does the procedure in the x-ray (radiology) department. The doctor, called a radiologist, uses x-rays and ultrasound scans to help them place the nephrostomy tube in the kidney. It can take 30 to 60 minutes.

Your doctor or nurse will put a fine tube called a [cannula](#) into a vein in your arm. They may use this to give you:

- a drip (infusion) of fluids
- antibiotics, to reduce your risk of getting an infection
- a sedative, to help you relax
- painkillers.

You lie flat on your tummy on an x-ray table. When you are comfortable, the doctor injects some local anaesthetic into the skin on the side of the back.

When the area is numb, the doctor gently puts a fine needle through the skin and into the kidney. They then put a guide wire through the needle. The doctor uses the guide wire to place the nephrostomy tube in the kidney. They use stitches or dressings keep the tube in place.

The nephrostomy tube may also have a locking system. This coils inside the kidney to keep the tube in place. The doctor connects the tube to a drainage bag outside the body, which collects the urine (pee).

You may feel discomfort when the doctor is putting the nephrostomy tube into the kidney. Tell them if you have any pain or feel anxious. They can give you a painkiller through the cannula in your arm.

After a nephrostomy

You are looked after in a hospital ward and usually need to stay in bed for a few hours. The nurses will check the nephrostomy tube is draining urine (pee). They will also check your blood pressure, pulse and temperature. Sometimes the doctor may do a test to check the position of the tube in the kidney. This is done by injecting a dye called a contrast into the tube. This shows up on an x-ray.

You may have some blood in your urine for about 2 days after the nephrostomy is put in. Blood in the urine is called haematuria. The area where the tube was put in may be uncomfortable. You will have regular painkillers. Any pain or discomfort usually gets better after about a week.

Possible complications from having a nephrostomy

The risk of complications from a nephrostomy is low. Possible complications include:

- infection
- bleeding from the kidney
- urine leaking from the kidney and collecting in the abdomen
- the nephrostomy tube falling out and needing to be placed again.

The doctors and nurses will check you regularly for any problems. This means if you have any, they can be treated straight away.

Before you go home

While you are in hospital, the nurses check the tube and empty the bag regularly. They will show you how to look after your nephrostomy. They can show you how to empty and change the drainage bag and will change the dressing. They will also tell you how long you need to wait before having a bath or shower.

You may decide to change the dressings yourself. Your nurse can show you how to do this. It may help to have a family member or close friend with you, so they can help when you get home.

The nurses in the hospital will arrange for you to see a practice nurse at your GP surgery, or for a district nurse to visit you at home. The practice nurse or district nurse will check the tube and change your dressing. They will also make sure you have enough drainage bags.

Before you go home, the nurse will tell you who to contact if you are worried that something is wrong with the nephrostomy. If the tube comes out, or if it stops draining urine, contact them straight away. You should also contact them straight away if:

- there is blood in your urine
- your urine looks cloudy or smells strongly
- it is painful when you pass urine in the normal way
- the area around the tube becomes uncomfortable, looks red or swollen, or feels warm.

These are all possible signs of an infection. You may need antibiotics.

Caring for a nephrostomy

Your nephrostomy tube will be at the side of the back. The tube is attached to a drainage bag. You usually wear the bag under your clothes and strap it to the thigh. Or you may have a bag that sticks to the skin directly over the tube.

It is important that you always keep the bag lower than the kidney. This helps the urine (pee) to drain. At night, you can attach a larger drainage bag that does not need emptying until the morning.

Once you are home, you will be able to go out as before. You should avoid sudden movements that may tug on the tube. At first, bending or climbing stairs may be uncomfortable. Place your hand over the nephrostomy when you need to move, in a way that feels uncomfortable. This can help to prevent

pulling on the tube. It might take you a little while to find the most comfortable position to sleep in.

It is natural to feel worried about coping with a nephrostomy at home. With help and support, people usually get used to it. Here are some important things to remember:

- Change the bag as often as your nurse or doctor has advised.
- Always wash your hands before and after you change the bag or the dressings.
- Drink plenty of fluids to help reduce the risk of infection. You should try to drink at least 2 litres (3½ pints) a day.

Your feelings about having a nephrostomy

Having a nephrostomy may affect the way you feel about your body ([body image](#)). Some people feel embarrassed and uncomfortable about other people seeing it. These feelings are a natural part of learning to cope with a nephrostomy. They usually get better over time.

If you find you are still having problems coping, ask your GP or your hospital doctor for help.

Everyone has their own way of dealing with difficult feelings. Some people find it helpful to talk to someone. This could be:

- a family member
- a friend
- your doctor
- your specialist nurse
- a trained counsellor – you can ask your doctor or nurse about this.

Joining our [Online Community](#) can be a good way to share experiences with people who understand what you are going through.

You can also contact our cancer support specialists for free on [0808 808 00 00](#) 7 days a week, 8am to 8pm for information and support.

We have more information about [getting help coping with difficult emotions](#).

About our information

References



Below is a sample of the sources used in our bladder cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Mottet N, Bellmunt J, Briers E, et al. Non-muscle-invasive bladder cancer (TaT1 and CIS). European Association of Urology (Internet), 2021. Available from uroweb.org/guideline/non-muscle-invasive-bladder-cancer (accessed September 2021).

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Ursula McGovern](#), Consultant Medical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

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Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.



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